

Python Assignment

Question 8 – Dictionary and Lambda Functions

Question: Define a dictionary **student** where roll number is the key and the value is another dictionary containing name, department and marks. The student dictionary has 5 records. Use lambda functions to perform:

- (i) Sort the dictionary according to the marks, high to low.
- (ii) Print the record of the student who scored maximum marks.
- (iii) Find the average marks of the students.
- (iv) Print the records of the students who scored more than the average marks.

Student Dictionary

```
student = {  
    1: {"name": "A", "dept": "CSE", "marks": 80},  
    2: {"name": "B", "dept": "IT", "marks": 90},  
    3: {"name": "C", "dept": "CSE", "marks": 70},  
    4: {"name": "D", "dept": "ECE", "marks": 85},  
    5: {"name": "E", "dept": "IT", "marks": 95}  
}
```

(i) Sort according to marks, high to low

```
result = sorted(student.items(),  
                 key = lambda x: x[1]["marks"],  
                 reverse = True)  
  
print(result)
```

(ii) Print the record of the student who scored maximum marks

```
result = max(student.items(),  
              key = lambda x: x[1]["marks"])  
  
print(result)
```

(iii) Find the average marks of the student

```
total = sum(x["marks"] for x in student.values())  
  
avg = total / len(student)  
  
print(avg)
```

(iv) Print the records of students who scored more than average marks

```
for a, b in student.items():  
    if b["marks"] > avg:  
        print(a, b)
```